NILANJAN MANDAL

Full-Stack Engineer | React, Next.js, React Native and Node.js

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PROFESSIONAL SUMMARY

Full-stack engineer with 2+ years of full-time experience and 2.5 years of internship experience building React Native apps, React admin panels and Node.js/TypeScript services. Delivered rewards, wallet, video-upload and payment workflows, and built a bounding-box annotation platform.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL
Frontend and mobile: React, Next.js, React Native, Expo, Tailwind CSS
Backend: Node.js, Express, REST APIs, Prisma, JWT, OAuth 2.0, RBAC, Zod
Data and jobs: PostgreSQL, Redis, BullMQ, WebSockets, Socket.IO
Cloud and testing: AWS (S3, Lambda, EC2), Docker, GitHub Actions, Jest, Supertest, Playwright

PROFESSIONAL EXPERIENCE

Member of Technical Staff Sep 2024 - Oct 2026
Tevis Software Private Limited (23 Ventures) | *Remote, India*

- Built React Native mobile interfaces and React admin panels, integrating Node.js/TypeScript services for rewards, campaigns, venues and wallets.
- Designed Express REST APIs and PostgreSQL schemas with Prisma for four branded mobile apps and a React admin console.
- Reduced p95 dashboard API response time from approximately 800 ms to under 150 ms by adding PostgreSQL indexes and tuning database queries.
- Implemented idempotency keys and transaction-safe wallet ledger entries, eliminating duplicate payouts observed during load testing.
- Moved settlement, notifications and bulk redemptions to BullMQ/Redis workers with retries, backoff and dead-letter handling for thousands of jobs per day.
- Scaled Socket.IO event delivery across Node.js cluster workers with a Redis adapter, supporting thousands of concurrent players in load tests.
- Secured write endpoints with JWT access tokens, refresh-token rotation, RBAC, Redis rate limiting and Zod validation.
- Containerised services with Docker and added GitHub Actions checks with Jest and Supertest integration tests against ephemeral PostgreSQL.

Software Engineering Intern Nov 2021 - May 2024
StepOut | *Remote, India*

- Rebuilt Node.js video ingestion with parallel, resumable S3 multipart uploads, increasing reliable upload size from approximately 500 MB to multi-gigabyte files.
- Integrated Razorpay order APIs, payment capture, webhook signature verification and idempotent reconciliation to keep payment records consistent with the gateway.
- Extended PostgreSQL models and migrations for video metadata and multilingual content; built AWS Lambda workflows with Playwright for PDF generation and document merging.

SELECTED PROJECTS

Flixr | YouTube publishing workflow automation

- Built a full-stack publishing workflow with a Next.js frontend, Node.js/Express APIs and PostgreSQL approval states. Integrated YouTube OAuth 2.0 and Redis/BullMQ upload queues with encrypted refresh tokens, quota-aware retries and persistent queue state.

HushMark | PII detection and redaction | hush-mark.nilanjan.xyz

- Built a React/TypeScript interface for text, document and audio redaction, including human review of low-confidence entities. Integrated OpenAI entity detection and Whisper transcription with asynchronous processing and Docker packaging.

EDUCATION

B.Tech in Information Technology | 2024 | CGPA: 8.37
RCC Institute of Information Technology, Kolkata